Plastic Pipes & Fittings

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
HDPE Pipe (Potable Water Supply)	IS 4984 (High Density Polyethylene Pipes for Water Supply — Specification); IS 4985:2021 (Unplasticized PVC Pipes for Potable Water Supplies — Specification); IS 15778:2007 (Chlorinated Polyvinyl Chloride (CPVC) Pipes for Potable Hot and Cold Water Distribution Supplies — Specification); IS 14333:1996 (High Density Polyethylene Pipes for Sewerage — Specification); IS 16088:2012 (Polyethylene Pipes for Irrigation Supply — Specification); IS 15225:2002 (Polyethylene Pipes for Water Supply — Polyethylene Materials and Compounds — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 15801:2008 (Polypropylene-Random Copolymer Pipes for Hot and Cold Water Installations — Specification); IS 14402:1996 (Glass Reinforced Plastics	ISO 4427-1 (General properties of PE piping for water supply networks); ISO 4427-2 (Requirements for extruded PE water supply pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 15874-1 (Baseline system aspects of PP piping for hot/cold water distribution); ISO 15874-2 (Specific manufacturing and physical requirements for PP hot/cold water pipes [2.1]); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings); ISO 15877-1 (Baseline system aspects of CPVC piping for hot/cold water installations); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	ASTM D3035 (Polyethylene Plastic Pipe for Water Pressure Applications); ASTM F714 (Polyethylene Plastic Pipe Based on Outside Diameter); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1598 (Time-to-Failure of Plastic Pipe Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings); ASTM D2444 (Impact Resistance of Thermoplastic Pipe and Fittings); ASTM D2837 (Hydrostatic Design Basis for Thermoplastic Pipe Materials or Pressure Design Basis for Thermoplastic Pipe Products)	DIN EN 12201-1 (Polyethylene Piping Systems for Water Supply — General); DIN EN 12201-2 (PE Pipes); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 1167-1 (Internal Pressure Resistance of Thermoplastics Pipes)	NSF/ANSI 61 (Drinking Water System Components); NSF/ANSI 14 (Plastic Piping Components); AWWA C901 (Polyethylene Pressure Pipes for Water Service)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	(GRP) Pipes, Joints and Fittings for Use for Water Supply, Sewerage and Industrial Purposes — Specification); IS 17546:2021 (Chlorinated Polyvinyl Chloride (CPVC) Fittings for Potable Hot and Cold Water Distribution Supplies — Specification); IS 10124 (Parts 1 to 13):1988 (Specification for Fabricated PVC Fittings for Potable Water Supplies); IS 8008 (Parts 1 to 7):2003 (Injection Moulded High Density Polyethylene (HDPE) Fittings for Potable Water Supplies — Specification); IS 17424:2021 (Polyethylene (PE) Electrofusion Fittings for Water Supply — Specification); IS 7634 (Parts 1 to 3) (Code of Practice for Plastics Pipe Work for Potable Water Supplies); IS 5382:1985 (Specification for Rubber Sealing Rings for Gas Mains, Water Mains and Sewers);IS 13592; IS 12231; IS 12818; IS 14182; IS 16087; IS 14885; IS 16098; IS 7834; IS 14735; IS 13593; IS 12235				
HDPE Pipe (Natural Gas Distribution)	IS 14885 (Polyethylene Pipes	ISO 4437-1 (Plastics Piping Systems for the Supply of Gaseous Fuels —	ASTM D2513 (Thermoplastic Gas Pressure Pipe, Tubing, and Fittings);	DIN EN 1555-1 (Plastics Piping Systems for Gas Supply —	AGA Standards (American Gas Association — Gas Distribution

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	for the Supply of Gaseous Fuels — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 16088; IS 15225; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	Polyethylene (PE) — General); ISO 4437-2 (Requirements for extruded PE gas supply pipelines); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems); ISO 4437-3 (Requirements for electrofusion and butt-fusion PE gas fittings); ISO 4437-4 (Requirements for valves integrated into PE gas line networks); ISO 15874-1 (Baseline system aspects of PP piping for hot/cold water distribution)	ASTM F2619 (Polyethylene Piping Systems for Natural Gas Distribution); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1598 (Time-to-Failure of Plastic Pipe Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	General); DIN EN 1555-2 (Polyethylene Pipes for Gas Supply); DIN EN ISO 1167-1 (Internal Pressure Resistance of Thermoplastics Pipes); DIN EN ISO 9080 (Hydrostatic Strength of Thermoplastics Pipes)	Systems); CSA B137 Series (Thermoplastic Pressure Piping); PHMSA Pipeline Safety Regulations (US Gas Pipeline Applications)
HDPE Pipe (Sewerage)	IS 14333 (High Density Polyethylene Pipes for Sewerage — Specification); IS 16087:2016 (Polyvinyl Chloride (PVC) Profile Wall Pipes for Non-Pressure Underground Sewerage and Drainage — Specification); IS 4984:2016 (High Density Polyethylene Pipes for Water Supply — Specification); IS 16088:2012 (Polyethylene Pipes for Irrigation Supply — Specification); IS 15225:2002 (Polyethylene Pipes for Water Supply — Polyethylene Materials and Compounds — Specification); IS 15265:2003 (Crosslinked	ISO 4427-1 (Plastics Piping Systems for Water Supply and for Drainage and Sewerage under Pressure — Polyethylene (PE)); ISO 4427-2 (Requirements for extruded PE water supply pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM F714 (Polyethylene Plastic Pipe Based on Outside Diameter); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM F2306 (12 to 60 in. Annular Corrugated Profile-Wall Polyethylene Pipe and Fittings for Gravity-Flow Sewer and Storm Sewer Applications); ASTM D2412 (External Loading Properties of Plastic Pipe by Parallel-Plate Loading); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN 12666-1 (Polyethylene Piping Systems for Non-Pressure Underground Drainage and Sewerage); DIN EN ISO 9969 (Ring Stiffness of Thermoplastics Pipes); DIN EN ISO 3127 (Impact Resistance of Thermoplastic Pipes); DIN EN ISO 527 (Tensile Properties of Plastics)	NSF/ANSI 14 (Plastic Piping Components); AWWA Sewer Applications Guidelines; EPA Wastewater Infrastructure Requirements

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 16098 (Part 1):2013 (Structured-Wall Plastics Piping System for Non-Pressure Drainage and Sewerage — Pipes and Fittings with Smooth External Surface, Type A — Specification); IS 16098 (Part 2):2013 (Structured-Wall Plastics Piping System for Non-Pressure Drainage and Sewerage — Pipes and Fittings with Non-Smooth External Surface, Type B — Specification); IS 14402:1996 (Glass Reinforced Plastics (GRP) Pipes, Joints and Fittings for Use for Water Supply, Sewerage and Industrial Purposes — Specification); IS 8008 (Parts 1 to 7):2003 (Injection Moulded High Density Polyethylene (HDPE) Fittings for Potable Water Supplies — Specification); IS 17424:2021 (Polyethylene (PE) Electrofusion Fittings for Water Supply — Specification); IS 7634 (Parts 1 to 3) (Code of Practice for Plastics Pipe				

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Work for Potable Water Supplies);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 14885; IS 15801; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 12235; IS 5382				
HDPE Pipe (Industrial Chemicals); HDPE Pipe (Mining Slurry)	IS 4984 (High Density Polyethylene Pipes for Water Supply — Specification); IS 14333 (High Density Polyethylene Pipes for Sewerage — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 14885; IS 16088; IS 15225; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	HDPE Pipe (Mining Slurry);ISO 15494 (Plastics Piping Systems for Industrial Applications — PE, PP and PVC-U); ISO 4427 (Polyethylene Piping Systems for Water Supply and Sewerage, where applicable); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM F714 (Polyethylene Plastic Pipe Based on Outside Diameter); ASTM D1598 (Time-to-Failure of Plastic Pipe Under Constant Internal Pressure); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings); ASTM D2412 (External Loading Properties of Plastic Pipe by Parallel-Plate Loading)	DIN EN ISO 15494 (Plastics Piping Systems for Industrial Applications — PE, PP, PVDF); DIN EN ISO 175 (Plastics — Determination of the Effects of Liquid Chemicals); DIN EN ISO 1167-1 (Thermoplastics Pipes, Fittings and Assemblies for the Conveyance of Fluids — Determination of Resistance to Internal Pressure); DIN EN ISO 527-1 (Tensile Properties of Plastics)	NSF/ANSI 14 (where applicable); ASME B31.3 (Process Piping Applications); Mining Industry Engineering Standards
HDPE Pipe (Cable Duct); HDPE Pipe (Telecommunication Duct)	IS 4984 (High Density Polyethylene Pipes for Water Supply — Specification); IS 14930 (HDPE Ducts for Optical Fibre Cable Installation); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS	HDPE Pipe (Telecommunication Duct);ISO 4427 (Polyethylene Piping Systems); ISO 15494 (Plastics Piping Systems for Industrial Applications); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM F2160 (Solid Wall High Density Polyethylene Conduit Based on Controlled Outside Diameter); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D2444 (Impact Resistance of Thermoplastic Pipe and Fittings)	DIN EN 61386-24 (Conduit Systems for Cable Management — Particular Requirements for Conduit Systems Buried Underground); DIN EN ISO 9969 (Thermoplastics Pipes — Determination of Ring Stiffness); DIN EN ISO 3127 (Thermoplastics Pipes — Determination of Impact Resistance); DIN EN ISO 527-1 (Tensile Properties of Plastics)	Telcordia GR Standards (Telecommunication Duct Applications); SCTE Standards (Telecom Infrastructure); Utility Owner Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 14333; IS 14885; IS 16088; IS 15225; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382				
HDPE Pipe (Drip Irrigation)	IS 9271; IS 14182:1994 (Unplasticized PVC Pipes for Irrigation — Specification); IS 4984:2016 (High Density Polyethylene Pipes for Water Supply — Specification); IS 14333:1996 (High Density Polyethylene Pipes for Sewerage — Specification); IS 16088:2012 (Polyethylene Pipes for Irrigation Supply — Specification); IS 15225:2002 (Polyethylene Pipes for Water Supply — Polyethylene Materials and Compounds — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 8008 (Parts 1 to 7):2003 (Injection Moulded High Density Polyethylene (HDPE) Fittings for Potable Water Supplies — Specification); IS 17424:2021	ISO 9261 (Agricultural Irrigation Equipment — Emitters and Emitting Pipe — Specification and Test Methods); ISO 4427-1 (General properties of PE piping for water supply networks); ISO 4427-2 (Requirements for extruded PE water supply pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM F714 (Polyethylene Plastic Pipe Based on Outside Diameter); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM F2620 (Heat Fusion Joining of Polyethylene Pipe and Fittings); ASTM D1598 (Time-to-Failure Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 9261 (Agricultural Irrigation Equipment — Emitters and Emitting Pipe); DIN EN ISO 4427-2 (Polyethylene Pipes for Water Supply); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 527 (Tensile Properties)	Irrigation Association (IA) Standards; EPA Water Efficiency Guidelines; ASABE Irrigation Standards

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	(Polyethylene (PE) Electrofusion Fittings for Water Supply — Specification); IS 7634 (Parts 1 to 3) (Code of Practice for Plastics Pipe Work for Potable Water Supplies);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 16087; IS 14885; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 12235; IS 5382				
HDPE Pipe (Sprinkler Irrigation)	IS 14151 (Part 1); IS 14151 (Part 2); IS 14182:1994 (Unplasticized PVC Pipes for Irrigation — Specification); IS 4984:2016 (High Density Polyethylene Pipes for Water Supply — Specification); IS 14333:1996 (High Density Polyethylene Pipes for Sewerage — Specification); IS 16088:2012 (Polyethylene Pipes for Irrigation Supply — Specification); IS 15225:2002 (Polyethylene Pipes for Water Supply — Polyethylene Materials and Compounds — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 8008 (Parts 1 to	ISO 8779 (Plastic Piping Systems for Irrigation — Polyethylene Pipes and Fittings); ISO 4427-1 (General properties of PE piping for water supply networks); ISO 4427-2 (Requirements for extruded PE water supply pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM F714 (Polyethylene Plastic Pipe Based on Outside Diameter); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings); ASTM F2620 (Heat Fusion Joining of Polyethylene Pipe and Fittings)	DIN EN ISO 4427-2 (Polyethylene Pipes for Water Supply); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 9080 (Hydrostatic Strength); DIN EN ISO 527 (Tensile Properties)	Irrigation Association Standards; ASABE Irrigation Equipment Standards; Agricultural Water Management Guidelines

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	7):2003 (Injection Moulded High Density Polyethylene (HDPE) Fittings for Potable Water Supplies — Specification); IS 17424:2021 (Polyethylene (PE) Electrofusion Fittings for Water Supply — Specification); IS 7634 (Parts 1 to 3) (Code of Practice for Plastics Pipe Work for Potable Water Supplies);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 16087; IS 14885; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 12235; IS 5382				
HDPE Pipe (Marine Outfall)	IS 4984 (High Density Polyethylene Pipes for Water Supply — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 14333; IS 14885; IS 16088; IS 15225; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 4427 (Polyethylene Piping Systems for Water Supply and Sewerage Applications); ISO 15494 (Industrial Applications); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM F714 (Polyethylene Plastic Pipe Based on Outside Diameter); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D2412 (External Loading Properties of Plastic Pipe); ASTM D1598 (Time-to-Failure Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 4427-2 (Polyethylene Pipes); DIN EN ISO 15494 (Industrial Applications); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 527 (Tensile Properties)	PIANC Guidelines (Marine Infrastructure); EPA Marine Discharge Guidelines; Coastal Engineering Standards
MDPE Pipe (Gas Distribution)	IS 14885 (Polyethylene Pipes for the Supply of	ISO 4437-1 (Plastics Piping Systems for the Supply of Gaseous Fuels — Polyethylene (PE)); ISO 4437-2	ASTM D2513 (Thermoplastic Gas Pressure Pipe, Tubing, and Fittings); ASTM F2619 (Polyethylene Piping	DIN EN 1555-1 (Plastics Piping Systems for Gas Supply); DIN EN 1555-2 (PE Pipes); DIN EN ISO	AGA Gas Distribution Standards; PHMSA Pipeline Safety Regulations; CSA B137 Series

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Gaseous Fuels — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	(Requirements for extruded PE gas supply pipelines); ISO 15874-1 (Baseline system aspects of PP piping for hot/cold water distribution)	Systems for Natural Gas Distribution); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1598 (Time-to-Failure Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	12162 (Thermoplastics Materials for Pipes and Fittings); DIN EN ISO 9080 (Hydrostatic Strength)	
LDPE Pipe (Agriculture)	IS 4985 (Unplasticized PVC Pipes for Potable Water Supplies — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 13593:1992 (UPVC Pipe Fittings for Agricultural Pumps — Specification);IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 8779 (Plastic Piping Systems for Irrigation — Polyethylene Pipes and Fittings); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM D2239 (Polyethylene Plastic Pipe Based on Controlled Inside Diameter); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 4427-2 (Polyethylene Pipes for Water Supply, where applicable); DIN EN ISO 9261 (Agricultural Irrigation Equipment); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 527 (Tensile Properties)	Irrigation Association Standards; ASABE Irrigation Standards; Agricultural Water Management Guidelines
PVC-U Pipe (Potable Water)	IS 4985 (Unplasticized PVC Pipes for Potable Water Supplies — Specification); IS 15778:2007 (Chlorinated Polyvinyl Chloride (CPVC) Pipes for Potable Hot and Cold Water Distribution	ISO 1452-1 (Plastics Piping Systems for Water Supply and for Buried and Above-ground Drainage and Sewerage under Pressure — PVC-U — General); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 4427-1 (General properties of PE piping for water supply networks); ISO 4427-2 (Requirements for extruded PE water supply pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5	ASTM D1785 (Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120); ASTM D2241 (Poly(Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series)); ASTM D1598 (Time-to-Failure of Plastic Pipe Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings); ASTM D2444 (Impact Resistance of Thermoplastic Pipe and Fittings); ASTM D2837	DIN EN ISO 1452-1 (PVC-U Piping Systems for Water Supply — General); DIN EN ISO 1452-2 (PVC-U Pipes); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 3127 (Impact Resistance)	NSF/ANSI 61 (Drinking Water System Components); NSF/ANSI 14 (Plastic Piping Components); AWWA C900/C905 (Water Distribution Applications)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Supplies — Specification); IS 4984:2016 (High Density Polyethylene Pipes for Water Supply — Specification); IS 15225:2002 (Polyethylene Pipes for Water Supply — Polyethylene Materials and Compounds — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 15801:2008 (Polypropylene-Random Copolymer Pipes for Hot and Cold Water Installations — Specification); IS 14402:1996 (Glass Reinforced Plastics (GRP) Pipes, Joints and Fittings for Use for Water Supply, Sewerage and Industrial Purposes — Specification); IS 17546:2021 (Chlorinated Polyvinyl Chloride (CPVC) Fittings for Potable Hot and Cold Water Distribution Supplies — Specification); IS 10124 (Parts 1 to 13):1988 (Specification for Fabricated PVC Fittings for Potable Water Supplies); IS 8008 (Parts 1 to 7):2003 (Injection Moulded High Density	(Parameters ensuring fitness for purpose of the jointed PE water systems); ISO 1452-3 (Design rules for injection-molded PVC-U pressure fittings); ISO 1452-4 (Direct operational specs for valves in PVC-U fluid pressure systems); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems); ISO 15877-1 (Baseline system aspects of CPVC piping for hot/cold water installations); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	(Hydrostatic Design Basis for Thermoplastic Pipe Materials or Pressure Design Basis for Thermoplastic Pipe Products)		

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Polyethylene (HDPE) Fittings for Potable Water Supplies — Specification); IS 17424:2021 (Polyethylene (PE) Electrofusion Fittings for Water Supply — Specification); IS 7634 (Parts 1 to 3) (Code of Practice for Plastics Pipe Work for Potable Water Supplies); IS 5382:1985 (Specification for Rubber Sealing Rings for Gas Mains, Water Mains and Sewers);IS 13592; IS 12231; IS 12818; IS 14182; IS 16087; IS 14333; IS 14885; IS 16088; IS 16098; IS 7834; IS 14735; IS 13593; IS 12235				
PVC-U Pipe (Sewerage)	IS 15328; IS 16087:2016 (Polyvinyl Chloride (PVC) Profile Wall Pipes for Non-Pressure Underground Sewerage and Drainage — Specification); IS 14333:1996 (High Density Polyethylene Pipes for Sewerage — Specification); IS 16098 (Part 1):2013 (Structured-Wall Plastics Piping System for Non-Pressure Drainage and Sewerage — Pipes and Fittings with Smooth External Surface, Type A —	ISO 1452-1 (General requirements for pressure-rated PVC-U piping); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems); ISO 1452-3 (Design rules for injection-molded PVC-U pressure fittings); ISO 1452-4 (Direct operational specs for valves in PVC-U fluid pressure systems)	ASTM D3034 (Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings); ASTM F679 (Large-Diameter Plastic Gravity Sewer Pipe and Fittings); ASTM D2412 (External Loading Properties of Plastic Pipe by Parallel-Plate Loading); ASTM D2321 (Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications)	DIN EN ISO 1452 (PVC-U Piping Systems, where applicable); DIN EN 1401-1 (PVC-U Solid Wall Pipes for Non-Pressure Underground Drainage and Sewerage); DIN EN ISO 9969 (Ring Stiffness of Thermoplastic Pipes); DIN EN ISO 3127 (Impact Resistance)	NSF/ANSI 14 (Plastic Piping Systems); NSF/ANSI 61 (Potable Water Applications, where applicable); AWWA Standards (Water/Sewer Applications)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Specification); IS 16098 (Part 2):2013 (Structured-Wall Plastics Piping System for Non-Pressure Drainage and Sewerage — Pipes and Fittings with Non-Smooth External Surface, Type B — Specification); IS 14402:1996 (Glass Reinforced Plastics (GRP) Pipes, Joints and Fittings for Use for Water Supply, Sewerage and Industrial Purposes — Specification); IS 7634 (Parts 1 to 3) (Code of Practice for Plastics Pipe Work for Potable Water Supplies);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 4984; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 5382				
PVC-U Pipe (Underground Drainage)	IS 15328; IS 16087:2016 (Polyvinyl Chloride (PVC) Profile Wall Pipes for Non-Pressure Underground Sewerage and Drainage — Specification); IS 16098 (Part 1):2013 (Structured-Wall Plastics Piping System for Non-Pressure Drainage and Sewerage — Pipes and Fittings	ISO 4435-1 (Plastics Piping Systems for Non-pressure Underground Drainage and Sewerage — Unplasticized Poly(vinyl chloride) (PVC-U) — General); ISO 4435-2 (Pipes); ISO 1452-1 (General requirements for pressure-rated PVC-U piping); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 1452-3 (Design rules for injection-molded PVC-U pressure fittings); ISO 1452-4 (Direct operational specs for valves in PVC-U fluid pressure systems); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems); ISO	ASTM D3034 (Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings); ASTM F679 (Large-Diameter Plastic Gravity Sewer Pipe and Fittings); ASTM D2412 (External Loading Properties of Plastic Pipe); ASTM D2321 (Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications)	DIN EN 1401-1 (PVC-U Solid Wall Pipes for Non-Pressure Underground Drainage and Sewerage); DIN EN ISO 9969 (Ring Stiffness); DIN EN ISO 3127 (Impact Resistance); DIN EN ISO 527 (Tensile Properties)	NSF/ANSI 14; EPA Wastewater Infrastructure Guidelines; Utility Authority Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	with Smooth External Surface, Type A — Specification); IS 16098 (Part 2):2013 (Structured-Wall Plastics Piping System for Non-Pressure Drainage and Sewerage — Pipes and Fittings with Non-Smooth External Surface, Type B — Specification); IS 14735:1999 (Unplasticized PVC Pipes and Fittings for Soil and Waste Discharge Systems Inside and Outside Buildings including Ventilation — Roof Drainage Fittings); IS 7634 (Parts 1 to 3) (Code of Practice for Plastics Pipe Work for Potable Water Supplies);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 14402; IS 7834; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 5382	9969 (Evaluation of ring stiffness on non-pressure underground conduit pipes)			
PVC Casing Pipe (Borewell)	IS 12818 (Unplasticized PVC Screen and Casing Pipes for Bore-Wells — Specification); IS 13592:2013 (Unplasticized PVC Pipes for Soil and Waste Discharge System Inside and Outside Buildings Including Ventilation —	ISO 1452-2 (Plastics Piping Systems for Water Supply and Buried/Above-ground Drainage and Sewerage under Pressure — PVC-U — Pipes); ISO 16486 (Plastics Piping Systems for Gaseous Fuels — PE, where applicable); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems); ISO 15877-1 (Baseline system aspects of CPVC piping for hot/cold water installations); ISO 15877-2 (Specific	ASTM F480 (Thermoplastic Well Casing Pipe and Couplings Made in Standard Dimension Ratios); ASTM D1785 (Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120); ASTM D2241 (Poly(Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series)); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings); ASTM D2444 (Impact Resistance of Thermoplastic Pipe and Fittings); ASTM D2837 (Hydrostatic Design Basis for	DIN EN ISO 1452-2 (PVC-U Pipes for Water Supply); DIN EN ISO 9969 (Ring Stiffness of Pipes); DIN EN ISO 3127 (Impact Resistance); DIN EN ISO 1167-1 (Internal Pressure Resistance)	NSF/ANSI 61 (Potable Water Applications, where applicable); National Ground Water Association Guidelines; Well Construction Authority Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Specification); IS 7834 (Parts 1 to 8):1987 (Specification for Injection Moulded PVC Fittings for Solvent Cement Jointing for PVC Piping Systems);IS 4985; IS 15778; IS 12231; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	Thermoplastic Pipe Materials or Pressure Design Basis for Thermoplastic Pipe Products)		
CPVC Pipe (Hot & Cold Water)	IS 15778 (Chlorinated Polyvinyl Chloride (CPVC) Pipes for Potable Hot and Cold Water Distribution Supplies — Specification); IS 13592:2013 (Unplasticized PVC Pipes for Soil and Waste Discharge System Inside and Outside Buildings Including Ventilation — Specification); IS 12818:2010 (Unplasticized PVC Screen and Casing Pipes for Bore-Wells — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 15801:2008 (Polypropylene-Random Copolymer	ISO 15877-1 (Plastics Piping Systems for Hot and Cold Water Installations — Chlorinated Poly(vinyl chloride) (PVC-C) — General); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems); ISO 15874-1 (Baseline system aspects of PP piping for hot/cold water distribution); ISO 15874-2 (Specific manufacturing and physical requirements for PP hot/cold water pipes [2.1]); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	ASTM D2846 (CPVC Plastic Hot- and Cold-Water Distribution Systems); ASTM F441 (CPVC Plastic Pipe); ASTM D1598 (Time-to-Failure Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings); ASTM D2444 (Impact Resistance of Thermoplastic Pipe and Fittings)	DIN EN ISO 15877-1 (CPVC Piping Systems for Hot and Cold Water); DIN EN ISO 15877-2 (CPVC Pipes); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 9080 (Hydrostatic Strength)	NSF/ANSI 61 (Drinking Water Components); NSF/ANSI 14 (Plastic Piping Systems); IAPMO UPC Certification Requirements

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Pipes for Hot and Cold Water Installations — Specification); IS 7834 (Parts 1 to 8):1987 (Specification for Injection Moulded PVC Fittings for Solvent Cement Jointing for PVC Piping Systems);IS 4985; IS 12231; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 16098; IS 14402; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382				
PPR Pipe (Hot & Cold Water)	IS 15801 (Polypropylene-Random Copolymer Pipes for Hot and Cold Water Installations — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 15874-1 (Plastics Piping Systems for Hot and Cold Water Installations — Polypropylene (PP) — General); ISO 15874-2 (Specific manufacturing and physical requirements for PP hot/cold water pipes [2.1]); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings); ISO 15877-1 (Baseline system aspects of CPVC piping for hot/cold water installations); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	ASTM F2389 (Pressure-rated Polypropylene (PP) Piping Systems); ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D1598 (Time-to-Failure Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings); ASTM D2657 (Heat Fusion Joining of Polyolefin Pipe and Fittings)	DIN EN ISO 15874-1 (Polypropylene Piping Systems for Hot and Cold Water); DIN EN ISO 15874-2 (PP Pipes); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 9080 (Long-Term Hydrostatic Strength)	NSF/ANSI 61; NSF/ANSI 14; CSA B137 Series (Thermoplastic Pressure Piping)
PB Pipe (Hot & Cold Water)	IS 15450 (Polybutylene (PB) Piping Systems for Hot and Cold Water Installations); IS 15265:2003 (Crosslinked	ISO 15876-1 (Plastics Piping Systems for Hot and Cold Water Installations — Polybutene (PB) — General); ISO 15876-2 (Pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for	ASTM D3309 (Polybutylene Plastic Hot- and Cold-Water Distribution Systems); ASTM D2662 (Polybutylene Plastic Pipe and Tubing); ASTM D1598 (Time-to-Failure Under Constant Internal Pressure); ASTM D1599 (Resistance	DIN EN ISO 15876-1 (Polybutene Piping Systems for Hot and Cold Water); DIN EN ISO 15876-2 (PB Pipes); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 9080 (Hydrostatic Strength)	NSF/ANSI 61; NSF/ANSI 14; IAPMO Plumbing Certification Requirements

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 15801:2008 (Polypropylene-Random Copolymer Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	purpose of the jointed PE water systems); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 15874-1 (Baseline system aspects of PP piping for hot/cold water distribution); ISO 15874-2 (Specific manufacturing and physical requirements for PP hot/cold water pipes [2.1]); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings); ISO 15877-1 (Baseline system aspects of CPVC piping for hot/cold water installations); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)		
PEX Pipe (Hot & Cold Water)	IS 15801 (Polypropylene-Random Copolymer Pipes for Hot and Cold Water Installations — Specification); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 15875-1 (Plastics Piping Systems for Hot and Cold Water Installations — Cross-linked Polyethylene (PE-X) — General); ISO 15875-2 (Pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 15874-1 (Baseline system aspects of PP piping for hot/cold water distribution); ISO 15874-2 (Specific manufacturing and physical requirements for PP hot/cold water pipes [2.1]); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings); ISO 15877-1 (Baseline system aspects of CPVC piping for hot/cold water installations); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	ASTM F876 (Crosslinked Polyethylene (PEX) Tubing for Pressure Applications); ASTM F877 (Crosslinked Polyethylene (PEX) Plastic Hot- and Cold-Water Distribution Systems); ASTM F2023 (Oxidative Resistance of Crosslinked Polyethylene (PEX) Tubing and Systems); ASTM D1598 (Time-to-Failure of Plastic Pipe Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 15875-1 (Crosslinked Polyethylene Piping Systems); DIN EN ISO 15875-2 (PEX Pipes); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 10147 (Degree of Crosslinking of PE-X)	NSF/ANSI 61; NSF/ANSI 14; PPI (Plastics Pipe Institute) PEX Guidelines; IAPMO UPC Certification
ABS Pipe (Industrial)	IS 17885 (ABS Piping Systems for Industrial Applications); IS	ISO 15493 (Industrial Applications — Acrylonitrile-butadiene-styrene (ABS), Unplasticized Poly(vinyl chloride) (PVC-U) and Chlorinated Poly(vinyl	ASTM D1527 (Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe, Schedules 40 and 80); ASTM D3965 (Rigid ABS Materials for Pipe	DIN EN ISO 15493 (Industrial Thermoplastics Piping Systems — ABS, PVC-U, PVC-C); DIN EN ISO 1167-1 (Internal Pressure	NSF/ANSI 14 (Plastic Piping Components); UL 94 (Flammability, electrical applications); Chemical Industry

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	chloride) (PVC-C) — Specifications for Components and the System)	and Fittings); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	Resistance); DIN EN ISO 3127 (Impact Resistance); DIN EN ISO 527 (Tensile Properties)	Piping Specifications
PVDF Pipe (Chemical Process)	IS 17885 (Industrial Thermoplastic Piping Systems) (Chemical Process Applications); IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 15494 (Plastics Piping Systems for Industrial Applications — Polyethylene (PE), Polypropylene (PP), and Polyvinylidene Fluoride (PVDF))	ASTM F1673 (Polyvinylidene Fluoride (PVDF) Plastic Pipe and Fittings for Industrial Applications); ASTM D3222 (Unmodified Poly(Vinylidene Fluoride) Molding and Extrusion Materials); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 15494 (Industrial Plastics Piping Systems — PE, PP, PVDF); DIN EN ISO 12086-1 (Fluoropolymer Materials); DIN EN ISO 175 (Chemical Resistance); DIN EN ISO 1167-1 (Internal Pressure Resistance)	NSF/ANSI 61 (Chemical/Water Applications where applicable); ASME B31.3 (Process Piping); Semiconductor Industry Material Requirements
PP-H Pipe (Chemical Service)	IS 17885 (Industrial Thermoplastic Piping Systems) (Chemical Service Applications); IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 15494 (Plastics Piping Systems for Industrial Applications — PP and Other Thermoplastics)	ASTM F2389 (Pressure-rated Polypropylene (PP) Piping Systems); ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D543 (Resistance of Plastics to Chemical Reagents); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 15494 (Industrial Plastics Piping Systems); DIN EN ISO 1873-1 (Polypropylene Materials); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 175 (Chemical Resistance)	ASME B31.3 (Process Piping Applications); Chemical Process Industry Specifications; NSF/ANSI 61 (where applicable)
PP-RCT Pipe	IS 15801 (Polypropylene-Random Copolymer Pipes for Hot and	ISO 15874-1 (Polypropylene (PP) Piping Systems for Hot and Cold Water — General); ISO 15874-2 (Specific manufacturing and physical	ASTM F2389 (Pressure-rated Polypropylene (PP) Piping Systems); ASTM D4101 (Polypropylene Injection and Extrusion Materials);	DIN EN ISO 15874-1 (Polypropylene Piping Systems); DIN EN ISO 15874-2 (PP Pipes); DIN EN ISO 1167-1 (Internal	NSF/ANSI 14 (Plastic Piping Components); NSF/ANSI 61 (Drinking Water Components); CSA B137 Series (Thermoplastic

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 16098; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	requirements for PP hot/cold water pipes [2.1]); ISO 15874-5 (Fitness for Purpose)	ASTM D1598 (Time-to-Failure of Plastic Pipe Under Constant Internal Pressure); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	Pressure Resistance); DIN EN ISO 9080 (Long-Term Hydrostatic Strength)	Pressure Piping Systems)
Structured Wall PE Pipe	IS 16098 (Series); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15801; IS 14402; IS 7834; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382	ISO 21138-1 (Plastics Piping Systems for Non-pressure Underground Drainage and Sewerage — Structured-wall Piping Systems of PVC-U, PP and PE — General); ISO 21138-3 (PE Structured-wall Pipes); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems)	ASTM F894 (Polyethylene (PE) Large Diameter Profile Wall Sewer and Drain Pipe); ASTM D2412 (External Loading Properties of Plastic Pipe by Parallel-Plate Loading); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN 13476-1 (Structured-Wall Plastic Piping Systems for Underground Drainage and Sewerage); DIN EN 13476-2 (PE Structured Wall Pipes); DIN EN ISO 9969 (Ring Stiffness); DIN EN ISO 3127 (Impact Resistance)	AASHTO M294 (Corrugated Polyethylene Pipe, Highway Applications); FHWA Drainage Guidelines; Municipal Utility Specifications
Structured Wall PVC Pipe	IS 16098 (Structured Wall Unplasticized PVC (uPVC) Piping Systems for Sewerage and Drainage); IS 13592:2013 (Unplasticized PVC Pipes for Soil and Waste Discharge System Inside and Outside Buildings Including Ventilation — Specification); IS 12818:2010 (Unplasticized PVC Screen and Casing	ISO 21138-1 (Plastics Piping Systems for Non-pressure Underground Drainage and Sewerage — Structured-wall Piping Systems of PVC-U, PP and PE — General); ISO 21138-2 (PVC-U Structured-wall Pipes); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems); ISO 15877-1 (Baseline system aspects of CPVC piping for hot/cold water installations); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1]); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods)	ASTM F794 (Poly(Vinyl Chloride) (PVC) Profile Gravity Sewer Pipe and Fittings); ASTM F949 (Poly(Vinyl Chloride) Corrugated Sewer Pipe with a Smooth Interior and Fittings); ASTM D2412 (External Loading Properties of Plastic Pipe); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN 13476-1 (Structured-Wall Plastic Piping Systems); DIN EN 13476-2 (PVC-U Structured Wall Pipes); DIN EN ISO 9969 (Ring Stiffness); DIN EN ISO 3127 (Impact Resistance)	AASHTO M304 (Profile Wall PVC Pipe Applications); FHWA Drainage Guidelines; Municipal Sewer Authority Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	Pipes for Bore-Wells — Specification); IS 7834 (Parts 1 to 8):1987 (Specification for Injection Moulded PVC Fittings for Solvent Cement Jointing for PVC Piping Systems);IS 4985; IS 15778; IS 12231; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 14402; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 12235; IS 7634; IS 5382				
Plastic Pipe Fittings (PE)	IS 8008 (Series); IS 15265:2003 (Crosslinked Polyethylene (PEX) Pipes for Hot and Cold Water Installations — Specification); IS 7834 (Parts 1 to 8):1987 (Specification for Injection Moulded PVC Fittings for Solvent Cement Jointing for PVC Piping Systems); IS 12235 (Parts 1 to 19):2004 (Thermoplastics Pipes and Fittings — Methods of Test);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15801; IS 16098; IS 14402; IS 14735; IS 17546; IS 10124; IS 13593; IS 17424; IS 7634; IS 5382	ISO 4427-1 (Plastics Piping Systems for Water Supply and for Drainage and Sewerage under Pressure — Polyethylene (PE) — General); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 4427-5 (Parameters ensuring fitness for purpose of the jointed PE water systems); ISO 1452-3 (Design rules for injection-molded PVC-U pressure fittings); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings)	ASTM D3261 (Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing); ASTM F1055 (Electrofusion Type Polyethylene Fittings for Outside Diameter Controlled Polyethylene Pipe and Tubing); ASTM D3350 (Polyethylene Plastics Pipe and Fittings Materials); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN 12201-3 (Plastics Piping Systems for Water Supply and for Drainage and Sewerage Under Pressure — Polyethylene (PE) — Fittings); DIN EN 1555-3 (Plastics Piping Systems for Supply of Gaseous Fuels — PE Fittings); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 178 (Flexural Properties of Plastics)	NSF/ANSI 14; NSF/ANSI 61 (Potable Water Applications); PPI Technical Guidelines

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
Plastic Pipe Fittings (PVC-U)	IS 7834 (Series); IS 12235 (Parts 1 to 19):2004 (Thermoplastics Pipes and Fittings — Methods of Test);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 7634; IS 5382	ISO 1452-1 (Plastics Piping Systems for Water Supply and for Buried and Above-ground Drainage and Sewerage under Pressure — PVC-U — General); ISO 1452-3 (Design rules for injection-molded PVC-U pressure fittings); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 1452-2 (Solid-wall extruded PVC-U pressure pipes for water mains); ISO 1452-4 (Direct operational specs for valves in PVC-U fluid pressure systems); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems);	ASTM D2466 (Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40); ASTM D2467 (Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80); ASTM D1784 (Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated PVC Compounds); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 1452-3 (PVC-U Piping Systems for Water Supply — Fittings); DIN EN 1401-1 (PVC-U Sewer Pipes and Fittings); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 3127 (Impact Resistance of Thermoplastic Pipes)	NSF/ANSI 14; NSF/ANSI 61; IAPMO UPC Certification Requirements
Plastic Pipe Fittings (CPVC)	IS 15778 (Chlorinated Polyvinyl Chloride (CPVC) Pipes for Potable Hot and Cold Water Distribution Supplies — Specification); IS 13592:2013 (Unplasticized PVC Pipes for Soil and Waste Discharge System Inside and Outside Buildings Including Ventilation — Specification); IS 12818:2010 (Unplasticized PVC Screen and Casing Pipes for Bore-Wells — Specification); IS 7834 (Parts 1 to 8):1987 (Specification for Injection Moulded PVC Fittings for Solvent Cement Jointing for PVC Piping Systems); IS 12235 (Parts 1 to 19):2004 (Thermoplastics Pipes and Fittings	ISO 15877-1 (Plastics Piping Systems for Hot and Cold Water Installations — Chlorinated Poly(vinyl chloride) (PVC-C) — General); ISO 15877-3 (Dimensions and design limits for CPVC hot/cold water system fittingsGeneral Structural Testing Methods); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 1452-3 (Design rules for injection-molded PVC-U pressure fittings); ISO 1452-5 (System integration assembly fitness criteria for PVC-U pressure componentsPolypropylene (PP) & Chlorinated PVC (PVC-C) Systems); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings); ISO 15877-2 (Specific manufacturing and physical requirements for CPVC hot/cold water pipes [2.1])	ASTM F437 (Threaded CPVC Plastic Pipe Fittings, Schedule 80); ASTM F438 (Socket-Type CPVC Plastic Pipe Fittings, Schedule 40); ASTM F439 (Socket-Type CPVC Plastic Pipe Fittings, Schedule 80); ASTM D1784 (Rigid CPVC Compounds); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 15877-3 (CPVC Piping Systems — Fittings); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 3127 (Impact Resistance); DIN EN ISO 178 (Flexural Properties)	NSF/ANSI 14; NSF/ANSI 61; IAPMO UPC Certification Requirements

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	— Methods of Test);IS 4985; IS 12231; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 15801; IS 16098; IS 14402; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 7634; IS 5382				
Plastic Pipe Fittings (PPR)	IS 15801 (Polypropylene-Random Copolymer Pipes for Hot and Cold Water Installations — Specification); IS 7834 (Parts 1 to 8):1987 (Specification for Injection Moulded PVC Fittings for Solvent Cement Jointing for PVC Piping Systems); IS 12235 (Parts 1 to 19):2004 (Thermoplastics Pipes and Fittings — Methods of Test);IS 4985; IS 13592; IS 15778; IS 12231; IS 12818; IS 14182; IS 16087; IS 4984; IS 14333; IS 14885; IS 16088; IS 15225; IS 15265; IS 16098; IS 14402; IS 14735; IS 17546; IS 10124; IS 13593; IS 8008; IS 17424; IS 7634; IS 5382	ISO 15874-1 (Plastics Piping Systems for Hot and Cold Water Installations — Polypropylene (PP) — General); ISO 15874-3 (Dimensions and design limits for PP hot/cold water system fittings); ISO 4427-3 (Configuration rules for PE water pipeline fittings); ISO 1452-3 (Design rules for injection-molded PVC-U pressure fittings)	ASTM F2389 (Pressure-rated Polypropylene (PP) Piping Systems); ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D2657 (Heat Fusion Joining of Polyolefin Pipe and Fittings); ASTM D1599 (Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings)	DIN EN ISO 15874-3 (Plastics Piping Systems for Hot and Cold Water Installations — Polypropylene (PP) — Fittings); DIN EN ISO 15874-5 (Fitness for Purpose of the System); DIN EN ISO 1167-1 (Internal Pressure Resistance); DIN EN ISO 178 (Flexural Properties)	NSF/ANSI 14; NSF/ANSI 61; CSA B137 Series

Source field: Plastic and Plastic Products